

Review

Modeling of solar energy systems using artificial neural network: A comprehensive review

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ARTICLE INFO

Keywords:

Artificial neural networks
Solar energy
Solar collectors
Solar assisted heat pumps
Solar dryers
Solar stills
Photovoltaic/thermal systems

ABSTRACT

The development of different solar energy (SE) systems becomes one of the most important solutions to the problem of the rapid increase in energy demand. This may be achieved by optimizing the performance of solar-based devices under some operating conditions. Intelligent system-based techniques are used to optimize the performance of such systems. In present review, an attempt has been made to scrutinize the applications of artificial neural network (ANN) as an intelligent system-based method for optimizing and the prediction of different SE devices' performance, like solar collectors, solar assisted heat pumps, solar air and water heaters, photovoltaic/thermal (PV/T) systems, solar stills, solar cookers, and solar dryers. The commonly used artificial neural network types and architectures in literature, such as multilayer perceptron neural network, a neural network using wavelet transform, Elman neural network, and radial basis function, are also briefly discussed. Different statistical criteria that used to assess the performance of artificial neural network in modeling SE systems have been introduced. Previous studies have reported that artificial neural network is a useful technique to predict and optimize the performance of different solar energy devices. Important conclusions and suggestions for future research are also presented.

1. Introduction

Energy demand is growing worldwide due to the rapid population growth and the incredible evolution in industry (Can Şener et al., 2018). Renewable energy resources, which depend on natural resources to generate an infinite supply of energy that is sustainable and non-polluting, is a promising alternative to the conventional energy resources and have gained significant importance in the recent centuries to overcome the energy shortages (Guven and Sulun, 2017). Among all renewable resources of energy, such as wind (Alhmoud and Wang, 2018), tidal (Loisel et al., 2018), biomass (Barnes, 2015) and geothermal (Tomasini-Montenegro et al., 2017), SE have attracted the greatest attention in many engineering and industrial applications such as solar refrigeration (Chen et al., 2018), electricity generation (Jain et al., 2018), domestic space heating (Sharma et al., 2017), solar

chimney (Kasaeian et al., 2017), solar radiation prediction (Kasaeian et al., 2016), and water desalination (Sharshir et al., 2016).

There are many devices which utilize SE to do some useful work such as solar collectors, solar assisted heat pumps, solar air and water heaters, PV/T systems, solar stills, solar cookers, and solar dryers. Numerous experimental and theoretical studies have been performed to figure out how the different operating parameters affect each of the aforementioned devices. However, the experimental studies are constrained by the number of the conducted experiments which may require high cost and consume more time. Moreover, the experimental based results are suitable only for little combinations of the investigated parameters with a little number of levels for each parameter. Theoretical based results, on the other hand, are suitable only for simplified models of the practical devices under many simplifying assumptions. It is worth mentioning that, the key reason of using ANN

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<https://doi.org/10.1016/j.solener.2019.01.037>

Received 20 October 2018; Received in revised form 3 January 2019; Accepted 11 January 2019

Available online 25 January 2019

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Nomenclature*Abbreviations*

ANFIS	Adaptive Neuro-Fuzzy Inference System
ANN	Artificial Neural Network
BP	Back-Propagation
CC	Correlation Coefficient
COD	Coefficient of Determination
COV	Coefficient of Variance
CRM	Coefficient Of Residual Mass
EC	Efficiency Coefficient
EML	Extreme Machine Learning
ENN	Elman Neural Network
FFNN	Feed-Forward Neural Network
GA	Genetic Algorithm
GWO	Grey Wolf Optimization
LM	Levenberg–Marquardt
MAE	Mean Absolute Error
MLP	Multilayer Perceptron
MRE	Mean Relative Error
MSE	Mean Square Error
NN	Neural Network
OIMP	Overall Index of Model Performance
PCG	Pola-Ribiere Conjugate Gradient
PSO	Particle Swarm Optimization

PV/T	Photovoltaic/Thermal
RBF	Radial Basis Function
RMSE	Root Mean Square Error
RNN	Recurrent Network
SCA	Sine Cosine Algorithm
SCG	Scaled Conjugate Gradient
SE	Solar Energy
WN	Wavelet Networks
WNN	Wavelet Neural Networks

Symbols

b_k	bias
d_{max}	maximum observed value
d_{min}	minimum observed value
$\bar{d}$	average of observed values
t	translation coefficient
v_k	summation's output
w_{kj}	weight
x_j	input
y_k	output signal
$\bar{y}$	average of prediction values
λ	dilation coefficient
σ_j	neuron width
$\varphi(\cdot)$	activation function
$\psi(u)$	wavelet function

based models instead of analytical based methods is that the analytical methods are generally quite cumbersome (sometimes solving of high nonlinear partial differential equations is needed) (Bellos and Tzivanidis, 2018; Elsheikh et al., 2018; Salazar et al., 2017) and require many simplifying assumptions. However, ANN based methods have been proved to be a useful tool to model different engineering systems under real-world conditions without involving in solving complicated mathematical models. ANN, as an intelligent based technique which mimics the behavior of the human brain in dealing with different problems instead of solving complex mathematical models, is used as a black-box model with the ability to learn and find out the nonlinear relation between the system inputs and the system outputs. ANNs have the generalization capability as it can handle unseen data faster and simpler than other classical methods after a learning process using few measured data sets. Therefore, ANN-based methods have attracted the attention of scientists and researchers in different engineering and industrial disciplines, for instance, for modeling and identification of mechanical systems (Orlowska-Kowalska and Szabat, 2007; Singh et al., 2009). Many researchers have summarized the use of the ANNs in many applications of renewable energy systems (Kalogirou, 2001) such as solar radiation prediction (Bou-Rabee et al., 2017; Kashyap et al., 2015; Shaddel et al., 2016), grid-connected photovoltaic power system (Ferlito et al., 2017), wind energy (Karabacak and Cetin, 2014), geothermal systems (Liu et al., 2018), photovoltaic applications (Almonacid et al., 2017; Mellit and Kalogirou, 2008; Miloudi and Acheli, 2015; Yadav et al., 2018), solar dryers (Prakash et al., 2016), and solar collectors (Ghritlahre and Prasad, 2018a). However, to our best knowledge, there are no published reviews on the application of ANNs in SE systems. For these reasons, this paper sheds some light on the application of ANNs in modeling and optimization of different SE systems. The commonly used architectures of ANNs that have been used in modeling SE systems have been briefly discussed. Different statistical criteria that used to assess the performance of ANN in modeling SE systems have been introduced. Therefore, this review has long term scientific value for SE researchers and beginners.

This comprehensive review covers the following points:

- A brief discussion about the types of ANNs, such as multilayer perceptron neural network, neural networks using the wavelet, the function of the radial basis, and Elman neural network (ENN), used in different SE applications is introduced.
- Different activation functions types used in different ANNs as well as different standard statistical performance evaluation criteria used in the evaluation of ANNs performance are also briefly discussed.
- The application of ANNs in different solar devices, like solar collectors, solar assisted heat pumps, solar air and water heaters, photovoltaic/thermal (PV/T) systems, solar stills, solar cookers, and solar dryers, are summarized in detail.
- Conclusions and suggestions for future research are also presented.

2. Artificial neural networks

ANN is a processor which is widely parallel distributed and is composed of simple processing units called neurons (Haykin, 2009); which has a natural capability for storing and figuring out the experimental knowledge to be valid for use. ANN exhibits excellent characteristics such as high-speed information processing, mapping capabilities, fault tolerance, adaptively, generalization, and robustness. These characteristics make ANN a powerful and smart tool for modeling, prediction, and optimization of the performance of different engineering systems. Therefore, it has been applied to complex nonlinear engineering problems in different real-world applications with a significant reduction in cost and time (Elsheikh et al., 2013; Jani et al., 2017; Karabacak and Cetin, 2014; Kashyap et al., 2015; Siami-Irdemoosa and Dindarloo, 2015).

In general, the neuron model used in designing many ANN models consists of a group of connecting links called synapses each of them has its own weight w_{kj} (as in Fig. 1). This weight is multiplied by its own input x_j before summing all weighted inputs as well as an externally bias b_k which is responsible for lowering or increasing the summation's output v_k . Then an activation function $\varphi(\cdot)$ is applied to that output to decrease the amplitude range of the output signal y_k into a finite value; different types of activation functions are tabulated in Table 1. These sequences could be expressed in mathematical terms as follows

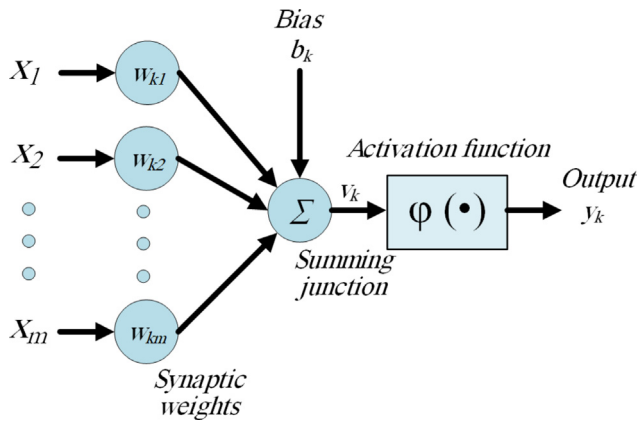


Fig. 1. Nonlinear model of a neuron.

(Haykin, 2009):

$$y_k = \varphi \left(\sum_{j=1}^m (w_{kj}) + b_k \right) \quad (1)$$

where k and j denote the neuron and synapse number, respectively.

2.1. Types of ANN

Different four models of the neural networks are discussed in this section including the Multilayer perceptron, the wavelet neural network, basis function which is radial, and the Elman neural network.

2.1.1. Multilayer perceptron neural network

The basic concepts of the Multilayer Perceptron (MLP) neural network will be discussed here. This kind of ANNs, in general, consists of one input layer, several hidden layers, and one output layer (Atkinson and Tatnall, 1997; Yan et al., 2006), and this is shown in Fig. 2(a). The neurons of each layer are connected with the neurons of the following layer (i.e., the input neuron is connected with the hidden layer neuron which connected with the output neuron), but, the neurons of the same layer are not connected with each other. The input layer's neurons receive the data then pass the data into the next layers until reaching the output layer.

The mathematical definition of this problem can be formulated by considering that, in the input layer, there is a set of M neurons. The

k_1 -th neuron receives the input x_i and the output of the neuron O_{1,k_1} is given by (Haykin, 2009):

$$O_{1,k_1} = f \left(\sum_{m=0}^M w_{mk_1} x_m \right), \quad k_1 = 1, \dots, K_1 \quad (2)$$

Then this output can be used as an input to the next layer called a hidden one, and the output of any neuron in the hidden layer can be given by:

$$O_{h,k_h} = \varphi \left(\sum_{k_{h-1}=0}^{K_{h-1}} w_{k_{h-1},k_h} O_{h-1,k_{h-1}} \right), \quad k_h = 2, \dots, K_h, \quad h = 1, \dots, N_h \quad (3)$$

where $\varphi(\cdot)$ represents the activation function, w_{0,k_h} , $k_h = 1, \dots, K_h$ represent the biases of the input and the hidden layers, respectively. w_{k_m,k_h} represent the weights between the input and hidden layer neurons (hidden layer neurons). M and K_h represent the number of the inputs and h th hidden layer neurons, respectively. N_h is the number found in the hidden layers.

The output neuron is calculated by summing all weighted outputs of the last hidden layer neurons:

$$y_l = \sum_{k_N=0}^{K_N} w_{k_N,l} O_{N,k_N}, \quad l = 1, \dots, L \quad (4)$$

where L is the number of the neurons used in the output layer, and, $w_{k_h,l}$, $k_h = 1, \dots, K_h$ are the weights of links connecting the neurons of the hidden and output layers.

The value of weights has an effect on the final output, therefore, to find these values the back-propagation (BP) learning (Ko and Lee, 2013) is used. However, the BP method is time consuming since it needs an iterative training process. In addition, the BP is also highly probably trapped in local minima.

2.1.2. Wavelet neural network

Wavelet neural networks (WNN) or wavelet networks (WN) make a hybrid between the theory of wavelets and the neural networks (Zhang and Benveniste, 1992). The WNNs inherit the advantages of the neural network and the wavelet transformation. In general, a WNN is made up of a feed-forward neural network (FFNN), and one hidden layer, where its activation functions can be generated from the orthonormal wavelet family (Zhang, 2009). These wavelet neurons can be called wavelons.

The simplest structure of the WNN consists of one input and one

Table 1
Different activation functions (Haykin, 2009).

Type	Description	Formula	Graph
Linear	The simplest activation function in which the output is proportional to the input	$\varphi(v_k) = cv_k$ where c represents the slope of the output of the function	
Sigmoid	A family of S-shaped functions.	$\varphi(v_k) = \frac{1}{1 + e^{-v_k}}$	
Gaussian	It is a continuous bell-shaped curve, has a peak at $v_k = 0$ and diminishes at the tails. The output node (high/low) is interpreted according to the membership of the class (1/0), based on how close the input of the net to a value of the average which has been selected.	$\varphi(v_k) = e^{-v_k^2}$	
Gaussian complement	It is the inversion version of the Gaussian function that is continuous, diminishes at $v_k = 0$, and has peaks at the tails.	$\varphi(v_k) = 1 - e^{-v_k^2}$	

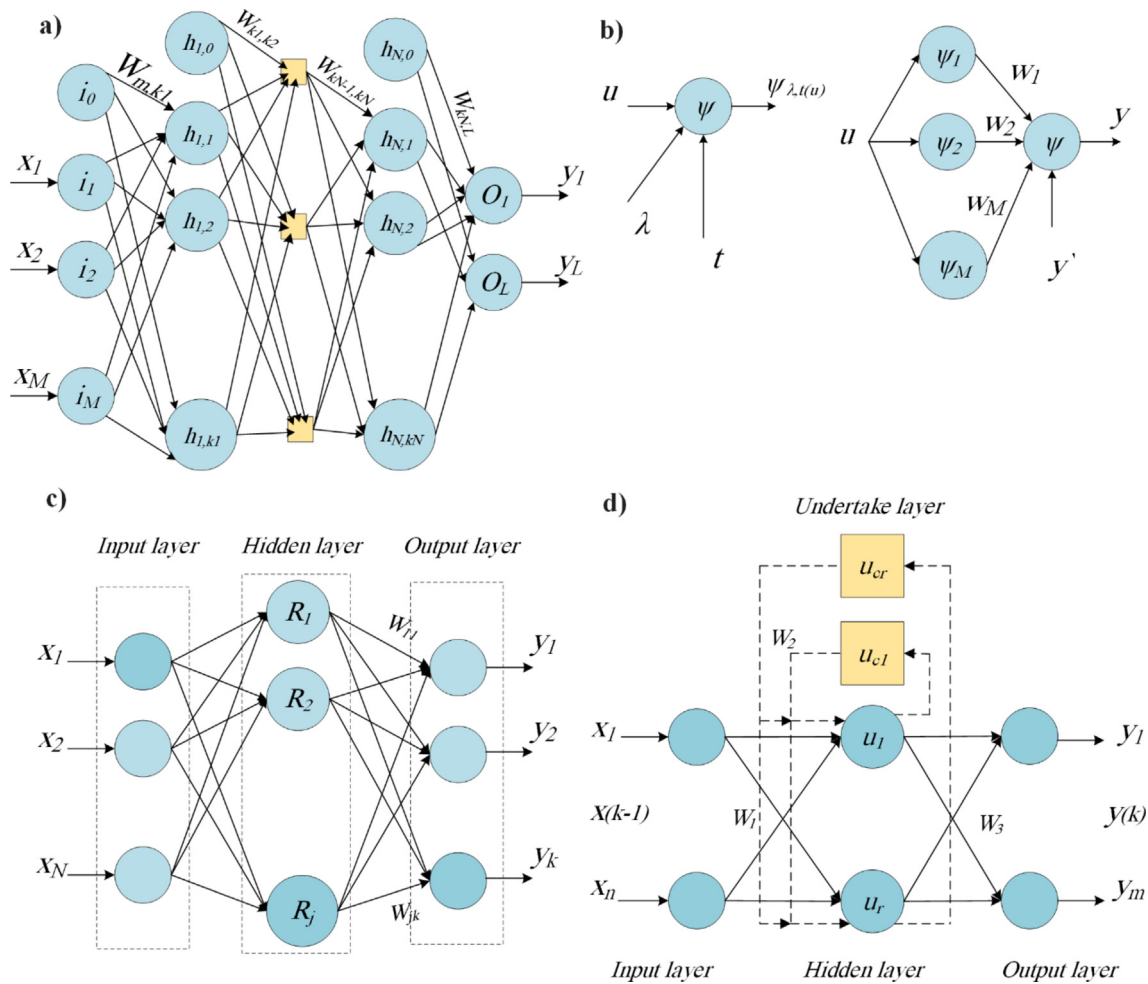


Fig. 2. The structure of (a) MLP, (b) Wavelet neural networks, (c) RBF (d) ENNs.

output as in Fig. 2(b left). The WNN's hidden layer is composed of wavelons, where its input coefficients contain the wavelet dilation and the translation parameters. The wavelons of the hidden layer lead to an output which not equal zero as the input is in a small area of the input space. The output of a WNN can be represented as a linear combination of the wavelet activation functions and this combination also is weighted.

The output can be represented as (Wang et al., 2013):

$$\psi_{\lambda,t}(u) = \psi\left(\frac{u-t}{\lambda}\right) \quad (5)$$

where t and λ represent the translation and the dilation coefficients respectively.

The general framework of a WNN which has one input one output is shown in Fig. 2(b right). The hidden layer of the WNN is composed of wavelons whose number is M . The output of the network is calculated by summing the weighted outputs of the wavelon (Zhao et al., 1998).

$$y(u) = \sum_{i=1}^M w_i \psi_{\lambda_i, t_i}(u) + \bar{y} \quad (6)$$

The $\bar{y}$ value deals only with the functions which have nonzero mean, where the wavelet function $\psi(u)$ also has a mean equals zero. The value of $\bar{y}$ is instead of the function $\phi(u)$, at the highest scale, in terms of multiresolution analysis of the wavelet. Through a WNN, all the coefficients $\bar{y}$, w_i , t_i , λ_i are learned by a predefined algorithm.

2.1.3. Radial basis function

Radial Basis Function (RBF) neural network is one of the most

efficient and powerful NNs, where it is composed of three main categories of layers (input layer, some hidden layers, and output layer) similar to MLP (as shown in Fig. 2(c)) (Broomhead and Lowe, 1988; Wu et al., 2010). Where the input layer in the RBF receives the input data and without computing power applies only to pass it through the hidden layer. Thereafter, the hidden layer performs some operation of the received data and extracted relevant information and send it to the output layer that construct the output data. However, the main difference between both of them (RBF and MLP) is that: the use of the RBF is done through the hidden layer, and it can be defined by the Gaussian function which is represented as (Broomhead and Lowe, 1988):

$$a_j(x) = \exp\left(-\frac{x_i - c_j}{2\sigma_j^2}\right) \quad (7)$$

where σ_j is the neuron width with the number j th, and x_i and c_j are the input and the center of RBF unit, respectively. Also, there exists some other functions can be used as in the following:

$$a_j(x) = \exp\left(-\left(\frac{x}{\sigma}\right)^2\right) \quad (8)$$

$$a_j(x) = \exp\left(\frac{1}{(\sigma^2 + x^2)^\alpha}\right), \quad \alpha > 0 \quad (9)$$

$$a_j(x) = (\alpha^2 + x^2)^\beta, \quad \alpha < \beta < 1 \quad (10)$$

The second difference is the output layer used the linear function that represented as a weighted sum of all the radial basis functions, and

it is defined as:

$$y_k = \sum_{j=1}^J w_{jk} R_j(x), \quad k = 1, 2, \dots, K \quad (11)$$

where K represents outputs' number; w_{jk} represents the weight which connect the j -th node of the hidden layer and the k -th node of the output layer; j is the number of nodes through the hidden layer.

2.1.4. Elman neural network

In general, the Elman neural network (ENN) is a recurrent network (RNN), which is a type of the ANNs where connections among the neurons create a directed cycle which in turn leads to a dynamic temporal behavior for it. The RNNs are different from the feedforward neural networks where they use the inside memory to treat the sequences of the inputs. This, in turn, helps them to do some applications like the unsegmented handwriting recognition or speech recognition.

The ENNs are proposed by Elman in (Elman, 1990). In ENNs, the outputs obtained from the hidden layer can be fed back onto themselves by using a recurrent layer, where the number of recurrent neurons is equal to the number of the hidden neurons. This gives the ENNs high ability to learn and construct a temporal and spatial pattern also, have a dynamic function, which gives the system the capability to be applicable with the time-varying properties. Where each node found in the hidden layer is connected with only one recurrent node using a fixed weight value.

Fig. 2(d) shows the general ENNs' structure in which it is composed of four main computing layers (input, hidden, recurrent, and output). Suppose that the input layer and the output layer consist of N and M , respectively, also, the hidden neurons number is N_h .

The input of the neural network is given by $u(k-1)$, while $x(k)$ and $x_c(k)$ represent the output of the hidden layer and recurrent layer, respectively, which defined as (Elman, 1990):

$$x(k) = f(w_2 x_c(k) + w_1 u(k-1)), \quad x_c(k) = x(k-1) \quad (12)$$

where w_1 is the weight of the connection between the input layer nodes and the hidden layer nodes and w_2 is the weight of the connection between the recurrent layer and the hidden layer. f represents the transfer function used in the hidden layer and it can be represented by the sigmoid function (see Table 1).

Thereafter, the output of the hidden layer is computed as the output of the neural network $y(k)$ which can be computed as (Elman, 1990):

$$y(k) = g(w_3 x(k)) \quad (13)$$

where w_3 represents the weight between the hidden layer and an output layer. g is the transfer function of the output layer.

The BP is used in ENNs to update the weights; where the error of network is computed as (Jia et al., 2014):

$$E = \sum_{k=1}^K (d(k) - y(k))^2 \quad (14)$$

where $d(k)$ represents the desired output of the input $u(k)$

2.2. Statistical performance evaluation criteria

In order to evaluate the quality of the prediction of neural networks, there are set of evaluation criteria such as mean square error (MSE), mean relative error (MRE), mean absolute error (MAE), root mean square error (RMSE), coefficient of variance (COV), correlation coefficient (CC), coefficient of determination (COD), efficiency coefficient (EC), overall index of model performance (OIMP), coefficient of residual mass (CRM) (Alazba et al., 2011; Rahman and Bala, 2010; Zangeneh et al., 2011). The definitions of these measures are given in Table 2, where n_s represents the number of observations, d and y are the desired (observed) and the prediction output values, respectively. d_{max} and d_{min} represent the maximum and the minimum observed value,

respectively. While $\bar{y}$ and $\bar{d}$ are the average of the prediction and observed values. The model which has smaller MSE, MRE, MAE, and RMSE value indicate the high quality of this algorithm and the value of these measure varies from 0, to ∞ where the best value is zero. In addition, the EC is used to assess the quality of the algorithm to fit between the predict and observed value and its optimal value is 1. The algorithm that has OI value of 1 indicates that the algorithm provides an ideal fit between the observed and predicted data. Meanwhile, when the CRM is zero this refers to the high performance of the model. Moreover, the R2 is used to determine the degree of similarity between the predict value and its observed value. In general, the best value for R2 is 1 which indicates the high performance of model.

3. ANN applications in SE

This section presents a comprehensive review of the published articles regarding the applications of ANNs in various SE devices, such as solar collectors, solar assisted heat pumps, solar air and water heaters, photovoltaic/thermal (PV/T) systems, solar stills, solar cookers, and solar dryers.

3.1. Solar collectors

Solar collectors are widely used devices to receive, collect, and concentrate (in some types) the sun radiation and transform it into useful heat used for heating up a fluid, water or air, which has many household or industrial applications depending on the amount of the generated heat and the rise in the temperature (Azad, 2018; Elsheikh et al., 2018; Raj and Subudhi, 2018). There are many types of solar collectors such as flat-plate, evacuated tube, parabolic trough, compound parabolic, and parabolic dish. Due to its importance in different SE collection systems, solar collectors must be efficiently designed to achieve optimum performance of such systems. Different optimization techniques have been proposed to select the optimal operating parameters for different types of solar collectors (Chauhan et al., 2017; Farahat et al., 2009; Kicsiny, 2014; Moghadam et al., 2011). Among all techniques used in solar collectors optimization, intelligent systems based optimization (such as ANN) is considered as a useful tool for modeling, simulation, and control of such applications (Farkas et al., 2001; Ghritlahre and Prasad, 2018a; Lalot and Lecoecuche, 2003). The solar collectors may be classified into two main categories, namely,

Table 2

Definitions of performance measures (Alazba et al., 2011; Rahman and Bala, 2010; Zangeneh et al., 2011).

Measure	Formula
Mean square error (MSE)	$MSE = \frac{1}{n_s} \sum_{i=1}^{n_s} (d_i - y_i)^2$
Mean relative error (MRE)	$MRE = \frac{1}{n_s} \sum_{i=1}^{n_s} \frac{d_i - y_i}{d_i}$
Mean absolute error (MAE)	$MAE = \frac{1}{n_s} \sum_{i=1}^{n_s} d_i - y_i $
Root mean square error (RMSE)	$RMSE = \sqrt{\frac{1}{n_s} \sum_{i=1}^{n_s} (d_i - y_i)^2}$
Coefficient of variance (COV)	$Cov = \frac{RMSE}{\frac{1}{n_s} \sum_{i=1}^{n_s} y_i} \times 100$
Correlation coefficient (CC)	$Corr = \frac{n_s \sum_{i=1}^{n_s} d_i y_i - (\sum_{i=1}^{n_s} d_i)(\sum_{i=1}^{n_s} y_i)}{\left[n_s \sum_{i=1}^{n_s} d_i^2 - (\sum_{i=1}^{n_s} d_i)^2 \right] \left[n_s \sum_{i=1}^{n_s} y_i^2 - (\sum_{i=1}^{n_s} y_i)^2 \right]}$
Coefficient of determination (COD)	$R^2 = \frac{(\sum_{i=1}^{n_s} (d_i - \bar{d})(y_i - \bar{y}))^2}{\sum_{i=1}^{n_s} (d_i - \bar{d})^2 \times \sum_{i=1}^{n_s} (y_i - \bar{y})^2}$
Efficiency coefficient (EC)	$EC = 1 - \frac{\sum_{i=1}^{n_s} (d_i - y_i)^2}{\sum_{i=1}^{n_s} (d_i - \bar{d})^2}$
Overall index of model performance (OIMP)	$OI = \frac{1}{2} \left(1 - \left(\frac{RMSE}{d_{max} - d_{min}} \right) + EC \right)$
Coefficient of residual mass (CRM)	$CRM = \frac{\sum_{i=1}^{n_s} y_i - \sum_{i=1}^{n_s} d_i}{\sum_{i=1}^{n_s} d_i}$

non-concentrating and concentrating solar collectors. The most common used non-concentrating solar collector type is the flat-plate solar collector. While the parabolic trough solar collector and solar concentrator dish are the most common used concentrating solar collectors. Herein, the applications of ANN in both solar collector types will be discussed as follows:

3.1.1. Concentrating solar collectors

Farkas and Géczy-Víg (2003) introduced and analyzed MLP to model three different types of flat-plate solar collectors (water, air and latent heat storage) (Fig. 3(a) and (b)). The structure of ANN (Fig. 1(c)) consists of an input layer includes three inlet parameters (global solar radiation, ambient temperature, and working fluid inlet temperature) and an output layer of a single output neuron (outlet temperature of working fluid). A sensitivity study on the training algorithm, number of neurons, and the size of the ideal training data set has been performed to select the most appropriate ANN topology. Moreover, the step function for the layers' neurons was analyzed and selected; among all available step functions (log-sigmoid, tan-sigmoid, hard-limit, and linear functions) the tan-sigmoid function and linear step function showed the best performance for the first layer and the second layer, respectively. An average error of 0.9 °C between the measured and NN calculated outlet collector temperature has been achieved.

(Benli, 2013) developed a FFNN to predict thermal performances of solar air collectors with two different surface shapes (corrugated and trapeze). BP algorithm was used in the developed neural network. Among different feed- forward BP learning algorithms, namely Pola-Ribiere Conjugate Gradient (PCG), Scaled Conjugate Gradient (SCG) and Levenberg–Marquardt (LM), the later was used due to its stability, fast convergence, and efficiency compared to other methods. The input variables to the NN are ambient temperature, inlet air temperatures, outlet air temperatures, air mass flow, solar irradiance, surface temperature of the collector; while the output variable is the collector's thermal performance. (Lecoeuche and Lalot, 2005) predicted the daily performance of solar air collectors by applying ANNs. The inputs of the NN are the thermal heat loss coefficients and the solar radiation, the output of the NN is the temperature of outlet air. The difference

between the predicted and the measured values is always inferior to 0.5%. (Kalogirou, 2006), developed six separate ANN models to predict the standard performance of a solar collector in terms of wind conditions (wind or no-wind), the collector heat capacity, the incidence angle modifier coefficients, the collector stagnation temperature and the collector time constant. Six ANNs were developed due to different type of output required for each model. The inputs variables to the ANN are the collector type (normal or roof integrated flat-plate collector), absorber type (normal, rolled, clipped, cushion, braided clamping, roll bonded, and welded), coating type (black chrome/galvanic, cobalt/galvanic, nickel pickmented, metal carbide, black nickel, ... etc.), absorber material (steel, stainless steel, aluminum, copper, and copper/aluminum), absorber mass, fin fluid content, insulation type (wool, polyurethane, and combination of two insulators), glazing thickness, fluid volume, absorber area, operating pressure, and volume flow rate. It was observed that, the estimation's accuracy increased when more cases and data are added to the databases, which could be used to enhance the design of the solar collector without performing more experimental tests.

Sözen et al. (2008) obtained a new formula using ANN to assess the efficiency of flat-plate solar collectors which were modeled as the ANN's output. The inputs parameters/variables of the ANN are solar radiation, surface temperature in collector, azimuth angle, tilt angle, declination angle, date, and time. The logistic sigmoid transfer function was applied. All inputs were normalized in the (0,1) range. The maximum deviation between the predicted and the actual (measured) efficiency was 2.55%. Tomy et al. (2016) used ANN to simulate flat plate solar collector using nanofluid working fluid consisted of silver and water. The ANN input parameters are the inlet temperature, mass flow rate, and radiation heat flux; while the output parameters are thermal efficiency, outlet temperature and heat transfer coefficient. Xie et al. (2009) developed an ANN based approach to assess the thermal performance of solar collectors. The ANN input parameters are the solar intensity, ambient temperature, azimuth angle, declination angle and tilt angle; while the output parameters are the heating capacity and the efficiency of the collector. The performance of the ANN was assessed using COV, RMSE and CC.

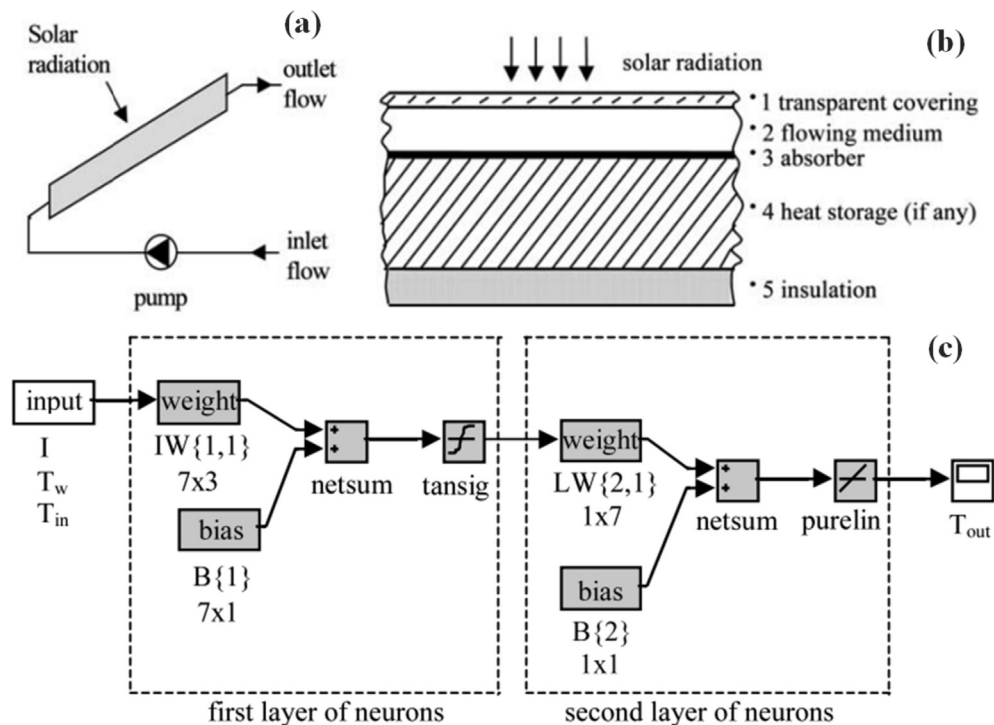


Fig. 3. (a) Flat-plate solar collector layout; (b) cross section; (c) ANN structure used for modeling the solar collector (Farkas and Géczy-Víg, 2003).

Caner et al. (2011) designed an ANN model to predict the thermal performance of two different types of solar air collector (flat and zig-zagged absorber surface). The input parameters of the proposed model were solar intensity, time of experiments, absorber model type, ambient and absorber surface temperature, outlet and inlet air temperatures, and stored water temperature; while the output was the collector efficiency. BP LM learning algorithms was used with hyperbolic tangent sigmoid and linear transfer functions applied to the hidden layer and the output layer, respectively. Statistical error analysis based on MAE, RMSE, the sum of squared errors of prediction, MRE, and R-squared was performed to assess the accuracy and reliability of the MLP model. Hamdan et al. (2016) developed an ANN model using resilient BP learning algorithm to analyze the heat transfer in an unglazed type flat-plate solar collector subjected to air passing behind the absorbing plate. The input parameters of the model were the total solar radiation incident on the collector surface, the mean temperature inside absorber plate surface, the mean temperature at the insulated surface of the collector, and the mean temperature of the absorber plate. Fischer et al. (2012) used ANN to determine the thermal performance of evacuated tubular collector as modeling this type of solar collectors using the state-of-the-art modeling mentioned in the European Standard EN 12975-2 experiences lack of accuracy. The input parameters to the ANN model were the incident angle of the beam irradiance, the beam, and diffuse irradiance, the temperature difference between ambient temperature and collector fluid inlet temperature and the mass flow rate; while the output was the collector output. The obtained results of the ANN model showed better accuracy compared with the state-of-the-art modeling.

Dikmen et al. (2014) used ANN, adaptive neuro-fuzzy inference system (ANFIS), and Genetic algorithm (GA) to predict and maximize the thermal performance of an evacuated tube solar collector shown in Fig. 4(a). LM algorithm was employed to train the ANN architectures. The inputs of the models were the solar radiation, ambient temperature, mean storage tank temperature, and tilt angle. Logistic sigmoid transfer function has been applied for both the hidden and output layers. ANN model used for the thermal performance prediction and the basic flow chart of the GA are shown in Fig. 4(a) and (b), respectively. The ANN and ANFIS model's performance were assessed using various standard statistical performance evaluation criteria such as RMSE, COV, and the coefficient of multiple determination. ANN method showed lower RMSE as well as COV and higher coefficient of multiple determination compared with ANFIS and hence the former had better performance and was recommended to model that type of solar collector.

3.1.2. Non-concentrating solar collectors:

Kalogirou et al. (1998) applied ANN to analyze the thermal performance of a solar steam generator connected to a parabolic collector. The input parameters of the ANN are ambient temperature, solar irradiance, time, effective aperture-area, flash-vessel dimensions (diameter,

height, and thickness), and flash-vessel content. Linear and logistic activation functions were used for the input and output layers, respectively; while Gaussian, Gaussian complement, and tanh activation functions were used in the hidden layer. BP algorithm was used to perform the learning process. ANN model shows better mapping capabilities compared to the analytical model. Facão et al. (2004) used two types of ANN (MLPs and RBFs) to model two different types of hybrid SE systems composed of solar collectors and heat pipes (tube or plate) to predict the thermal efficiency and useful heat gained. Two different types of models have been constructed, namely, MLP and RBF models. Energy balances equations were used to get a mathematical model of the system based on Monte Carlo simulation which used to generate the training data of the ANN models. MLPs showed better performance compared with RBFs based on MSE criterion. The inputs parameters for both ANN models are ambient temperature, solar radiation, inlet water temperature, inlet gas temperature, water and gas mass flow rates, condenser length, and evaporator length.

Recently, Heng et al. (2019) developed an innovative fast and accurate method to predict the thermal performance of a parabolic trough solar collector. This method is composed of an ANN model and superposition principle. The inputs of the models are heat flux ($0\text{--}60000\text{ W/m}^2$), fluid velocity ($0.35\text{--}1.45\text{ m/s}$), and time (1200 s with 5 s interval); while the output of the model is the rise in the exit temperature of the collector. The architecture of the ANN model consists of two hidden layers with 68 neurons in each one in addition to the input and output layers. The proposed method requires less than 1 min to predict the one-day exit fluid temperature.

Loni et al. (2018) applied an ANN technique (called group method of data handling ANN) to model the dependency of the cavity heat gain and the thermal efficiency of a hemispherical cavity receiver integrated with a solar dish concentrator on solar irradiation, ambient temperature, outlet temperature. The obtained R^2 values based on the propose model, were 0.9709 and 0.9567 for the cavity heat gain and the thermal efficiency, respectively; which reveals the high precision of the model. In another study, Loni et al. (2017) investigated the thermal performance of a parabolic dish concentrator with a rectangular-tubular cavity receiver using two different methods, namely, numerical modeling method and an ANNs based method. The thermal oil was used as the working fluid in the solar collector system. The inputs of the ANN model were solar irradiance, the cavity depth, and the diameter of tube; while the output was the thermal performance of the cavity receiver. The obtained results showed a good agreement with $R^2 = 0.99$ between the predicted results of ANN and the calculated numerical modeling results. Moreover, they reported that ANN based method can be used as an effective tool which saves the time and cost to determine the optimal structure of the cavity receiver compared to the investigated numerical method.

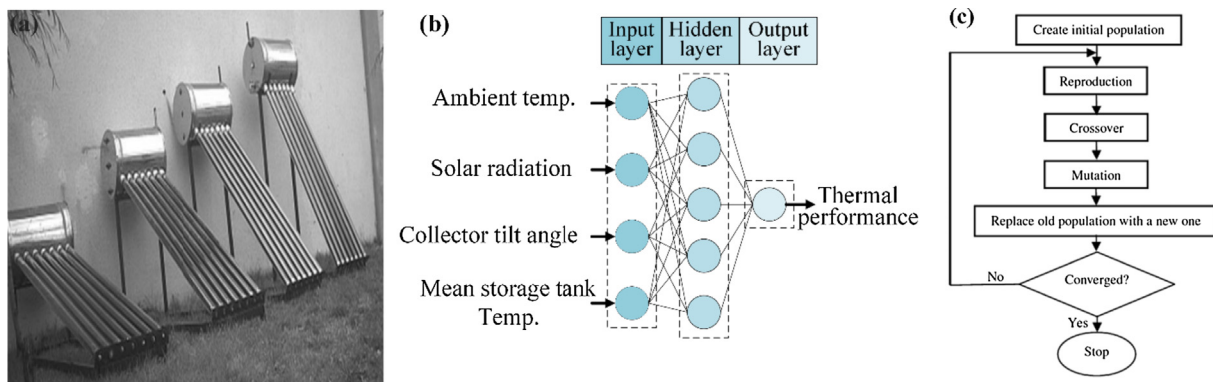


Fig. 4. (a) Evacuated tube solar collectors; (b) ANN model used for thermal performance prediction; (c) GA flow chart(Dikmen et al., 2014).

3.2. Solar air heaters

Solar air heater is a kind of heat exchanger used to produce warm air for both domestic and industrial applications (Abuşka, 2018; Cuzminschi et al., 2018; Ravi and Saini, 2016). Solar air heaters possess many advantages such as the inherent simplicity of the systems which require little maintenance, and have less severe leakage and corrosion problems compared to liquid based solar heaters; however they suffer from low thermal efficiency (Alam and Kim, 2017; Ghiami and Ghiami, 2017; Saxena et al., 2015). To overcome this drawback, optimization methods should be applied to select the optimal operating parameters to enhance the overall system efficiency (Hedayatzadeh et al., 2016; Moradi et al., 2017; Singh and Dhiman, 2016). One of the most efficient method used to perform this optimization is the intelligent based methods including ANN.

Esen et al. (2009) constructed MLP and WNN models to predict the outlet air temperature and thermal efficiency of a solar air heater shown in Fig. 5(a). The examined solar air heater consists of an absorber plate, collector box, glass cover, fan, and fan engine. A pyranometer was used to measure the solar irradiation on the absorber plate surface. Thermocouples were used to measure the temperature of the various components of the solar air heater. Anemometer was used to measure the wind speed. Different architectures of both models as well as training parameters were investigated and evaluated using several statistical validation criteria such as RMSE, coefficient of variation, and coefficient of multiple determinations. The input parameters for both models are solar radiation intensity, inlet air temperature, and temperatures at some locations (four locations) on the absorbing plate. Three learning algorithms were used in the MLP model (LM back-propagation, Pola–Ribiere conjugate gradient, and SCG); while the LMBP was used for WNN model Fig. 5(b). The used ANN model consists of an input layer, a wavelet layer, and an output layer. Two types of activation functions have been used, namely tangent sigmoid and Morlet wavelet, for MLP and WNN, respectively. The obtained results revealed that LM learning algorithm showed the best performance among other learning algorithms. However, WNN had better performance compared with MLP. Ghritlahre and Prasad (2017) constructed an ANN model to determine the thermal performance of a unidirectional flow solar air heater. The input parameters of the ANN are ambient temperature, air inlet temperature, mass flow rate, and solar radiation intensity; while the output parameters are temperature difference of air, heat gained by air, and thermal efficiency as shown in Fig. 5(c). Four different training functions, namely, SCG back-propagation, one-step secant back-propagation, Levenberg–Marquardt back-propagation, and conjugate gradient BP with Polak–Ribière updates, were compared using MSE and coefficient of multiple determinants to get the best performance. Levenberg–Marquardt BP training function showed the best performance among other training functions. Aidinlou and Nikbakht (2017) used ANN to predict the thermohydraulic performance of a solar air heater in terms of friction factor and Nusselt number. Velocity and temperature features were used as inputs to the ANN instead of geometric parameters which cause opportunity for comparison and investigation of the roughness based on the friction factor and the Nusselt number. Saravanakumar et al. (2013) used ANN for the prediction of the performance of solar air heater using different thermal storage materials. The input parameters of the ANN are solar radiation intensity, ambient temperature, and air velocity, while the output parameters are outlet air temperature and the first and second law efficiencies of the solar air heater. Omojaro et al. (2013) utilized ANN to model the efficiency of solar air heaters. The input parameters of the ANN were time, air inlet temperature, air outlet temperature, and temperature difference, while the ANN output was the overall efficiency. Varol et al. (2010) predicted the performance of a solar collector system working with heat transfer oil and sodium carbonate decahydrate phase change material as a working fluid using three different soft computing techniques, namely ANN, support vector machines and ANFIS. BP learning algorithm, with

linear activation function in the input and output layers and log-sigmoid activation function in the hidden layer, was used. The support vector machines model had the best performance followed by ANN and finally ANFIS for the case under study. Ghritlahre and Prasad (2018b) developed a feed forward ANN model to predict the heat transferred from roughened absorber plate to air passing through the ducts of solar air heater. Two different types of solar air heater were used in the experiments: the first has single sided-absorber plate and the second has three-sided absorber plate. The inputs of the model were number of rough surfaces side, roughness pitch, relative roughness size, relative roughness height, and Reynolds Number; while the output was Nusselt number. LM algorithm has been used to get the optimal architectures of the proposed models. The values of RMSE, MAE were 0.89202 and 0.66261, respectively during training stage. While, for testing stage these values were 0.55094 and 0.31683, respectively.

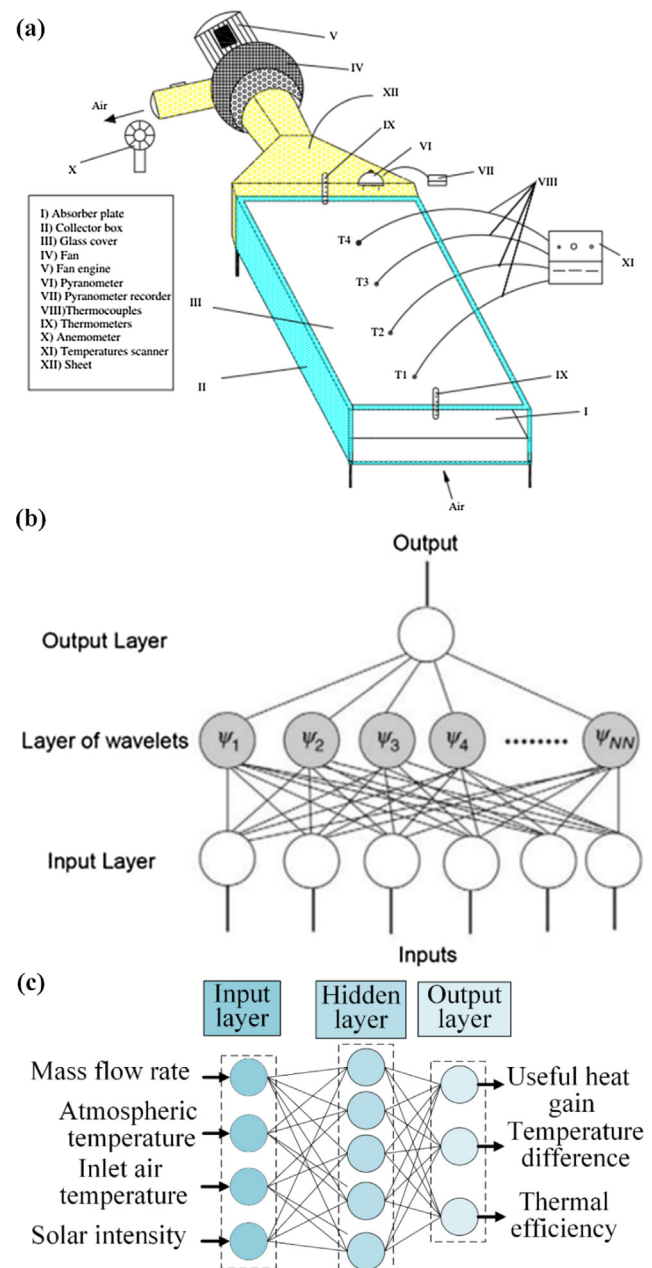


Fig. 5. (a) Schematic assembly of the solar air heater system (Esen et al., 2009); (b) Structure of WNN (Esen et al., 2009); (c) ANN model of solar air heater (Ghritlahre and Prasad, 2017).

3.3. Solar water heaters

A solar water heater is a thermal system (passive or active) used to convert the solar radiation into useful heat which is then transmitted to water by heat transfer phenomena or by a pumping system, this heated water is used in small-scale applications such as domestic purposes by storing it in an insulated tank (Benli, 2016; Mostafaeipour et al., 2017). It consists of a solar collector (in some cases multiple solar collectors), a storage tank and a circulation system for the working fluid used in transferring heat between the heater's components. Many investigations have been carried out to predict and assess the performance of the solar water heater systems and to select the optimal operating parameters to maximize their efficiencies (Assari et al., 2018; Hajabdollahi and Hajabdollahi, 2017; Jahangiri Mamouri and Bénard, 2018). ANNs, as an artificial intelligence-based tool, is used by many researchers for this purpose.

Kalogirou et al. (1999) presented an ANN model applied to solar domestic water heating systems to estimate the rise in the stored water temperature and the useful energy gained from the system. Thirty data sets were used to train the ANN by varying collector areas between 1.81 m and 4.38 m under a large variety of weather conditions. Both horizontal and vertical storage tanks with open and closed systems have been considered. Among all network architectures, that have been investigated to get the best overall performance, feed-forward architecture with eight neurons in the input layer, three slabs in the hidden layer (18 neurons for each one), and two neurons in the output layer was the best. The BP learning algorithm has been used. The ANN architecture as well as the used activation functions for each neuron set are shown in Fig. 6(a). The developed ANN architecture is composed of three main layers. The first layer is the input layer and consists of a single slab contains eight neurons with linear activation function. The second layer is the hidden layer and consists of three slabs, each slab contains 18 neurons. Three different activation functions were used in the hidden layer for each slab, namely, Gaussian activation function, tanh activation function, and Gaussian complement activation function. The third layer is the output layer and consists of a single slab contains two neurons with logistic activation function. The input parameters are the collector area, tank type (horizontal or vertical), mean storage tank heat loss coefficient, storage capacity, system type (open or closed), solar irradiation, mean ambient temperature, and mean temperature of cold water. The output parameters are the temperature rise of the water and the useful energy gained from the system. The accuracy of the predictions was within 7% to 9%. Later, Kalogirou and Panteliou (2000) developed an ANN model to predict the long-term system performance of a domestic thermosiphon type solar water heating system. Two different structures of ANNs models have been trained using the monthly data generated from a modeling program based on ISO 9459-2

standards. The ANNs were applied to estimate the SE output of the system for identical draw-off and storage tank capacity and for demand temperatures of 35 and 40°C for the first and the second NNs, respectively. The ANN architecture as well as the used activation functions for each neurons set are shown in Fig. 6(b). The developed ANN architecture is composed of three main layers. The first layer is the input layer and consists of a single slab contains two NNs with linear activation function; the first NN has 15 neurons while the second NN has 16 neurons. The second layer is the hidden layer and consists of three slabs, each slab contains 8 neurons. Three different activation functions were used in the hidden layer for each slab, namely, Gaussian activation function, tanh activation function, and Gaussian complement activation function. The third layer is the output layer and consists of a single slab contains two NNs with logistic activation function; the first NN has one neuron while the second NN has two neurons. The input parameters are the month, daily solar irradiance, mean ambient temperature, mean cold water temperature, specific volume of the system, system type (open or closed), tank type (horizontal or vertical), heat loss coefficient of the storage tank, and the maximum temperature of the draw-off water. The output parameter is the value of the SE output of the system.

Cetiner et al. (2005) established an ANN model to predict the performance (thermodynamic efficiency) of solar water heater system consists of an absorber, a cylindrical concentrator, a water store, a heat exchanger, a pump and a control unit. The ANN inputs parameters are solar radiation intensity, ambient temperature, inlet temperature into the absorber, and wind velocity. Yaïci and Entchev (2014) predicted the performance of a solar-based domestic hot water and space heating systems using ANNs. Various system performance parameters, such as the heat transferred from the solar collector to the heat exchanger, the preheat tank stratification temperatures, the heat input to the auxiliary propane-fired tank, were predicted using the proposed NN model. The input variables are the solar radiation on the horizontal and inclined planes, the ambient air temperature, the preheat tank stratification temperatures. The used ANN is MLP with an input layer, one hidden layer (20 hidden neurons), and an output layer was trained by applying two different learning algorithms, namely the SCG and the Levenberge-Marquardt. The later has proven to be flexible and robust with fast optimization compared to the later. Souliotis et al. (2009) developed a group method of data handling ANN model integrated with the well-known commercial energy simulation software TRNSYS to determine the performance of a hybrid collector storage consists of solar water heater with cylindrical water storage tank inside a compound parabolic concentrator reflector trough. The input parameters of the model were month, total radiation, ambient air temperature, wind speed, and, incident angle; while the output of the model was the mean storage tank temperature. The ANN model is used through the Excel interface in

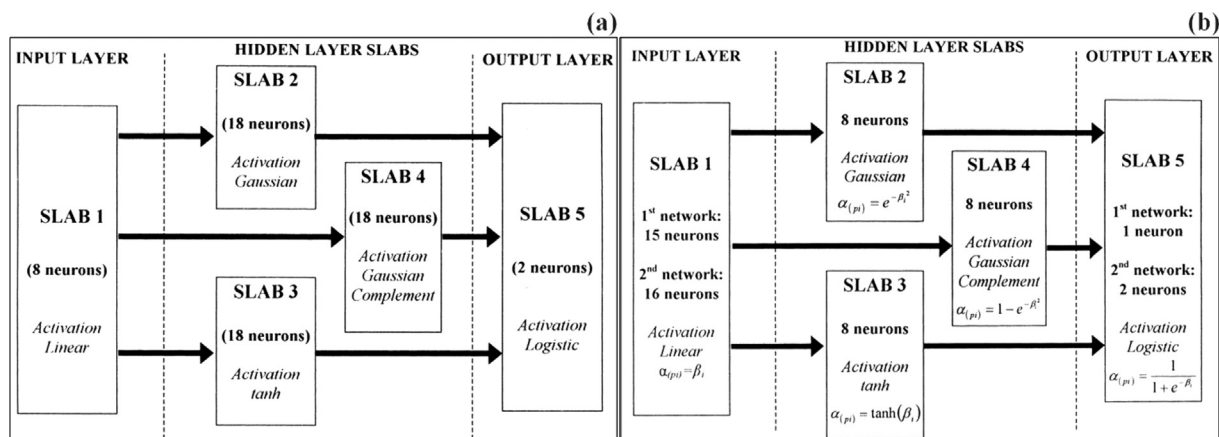


Fig. 6. Neural network architecture employed: (a) (Kalogirou et al., 1999); (b) (Kalogirou and Panteliou, 2000).

TRNSYS to predict the annual performance of the system using values of a typical meteorological year, which enables the combination between the two approaches, namely, black box modeling capabilities of the ANN and radiation and power modeling capabilities of TRNSYS.

3.4. Solar assisted heat pumps

Heat pumps, in general, has ability to deliver heat energy more than the work input so they are classified as energy efficient devices (Mohanraj et al., 2018b). One more advantage of heat pumps is their capability to integrate with different renewable energy sources such as geothermal and solar resources which significantly enhances their performance. However, integrating with SE resources is less expensive than integrating with the geothermal energy resources due to the excavation of the heat exchangers used in the later. Heat pumps are used in many engineering applications such as domestic heating, green house heating, drying, and water desalination applications (Mohanraj et al., 2018a). Therefore, due to their importance, many optimization investigations have been carried out to select optimal operating parameters (Calise et al., 2016; Li and Kao, 2018; Poppi et al., 2018). ANN is one of the optimization tools used for this purpose as discussed in the following paragraphs.

Mohanraj et al. (2008), Mohanraj et al. (2009b) constructed an ANN model to determine the performance of a solar assisted heat pump (Fig. 7(a)) consists of an evaporator (glazed solar collector), air cooled condenser, hermetically sealed compressor (R22), liquid receiver, refrigerant drier, and thermostatic expansion device. The ANN input is

the solar radiation intensity; while the outputs are SE input ratio, coefficient of performance, air temperature at condenser outlet, and compressor pressure ratio (Fig. 7(b)). A feed forward network, with one hidden layer and Levenberg Marquardt learning algorithm with a logistic sigmoid transfer function for the both hidden and output layers, was used. In another study, the same authors (Mohanraj et al., 2009a) developed another ANN model to investigate exergy of the same solar-assisted heat pump system. Exergy efficiency and exergy destruction of each component of the system were predicted using ANN based on measured meteorological parameters (solar intensity and ambient temperature). Fig. 7(c) illustrates the ANN architecture used to predict the exergy destruction of the system components; the same architecture was used for the exergy efficiency of each component by replacing destruction with efficiency and by updating the number and the weights of the neurons. The ANN architecture is composed of three layers. The first is the input layer which consists of two neurons represent the air temperature and the solar intensity. The second layer is the hidden layer. The third is the output layer which consists of six neurons represent the energy destruction of various components of the solar assisted heat pump including compressor, condenser, expansion valve, and solar collector in addition to the destruction of the overall system. In another work, Kai et al. (2009) introduced RBF ANN model to analyze the thermal performance of a solar assisted heat pump. Kumar et al. (2016) introduced ANN model integrated with GA to predict and optimize the performance of a solar assisted heat pump. The input parameters of ANN model are solar intensity and ambient temperature, while the outputs are heating capacity, power consumption,

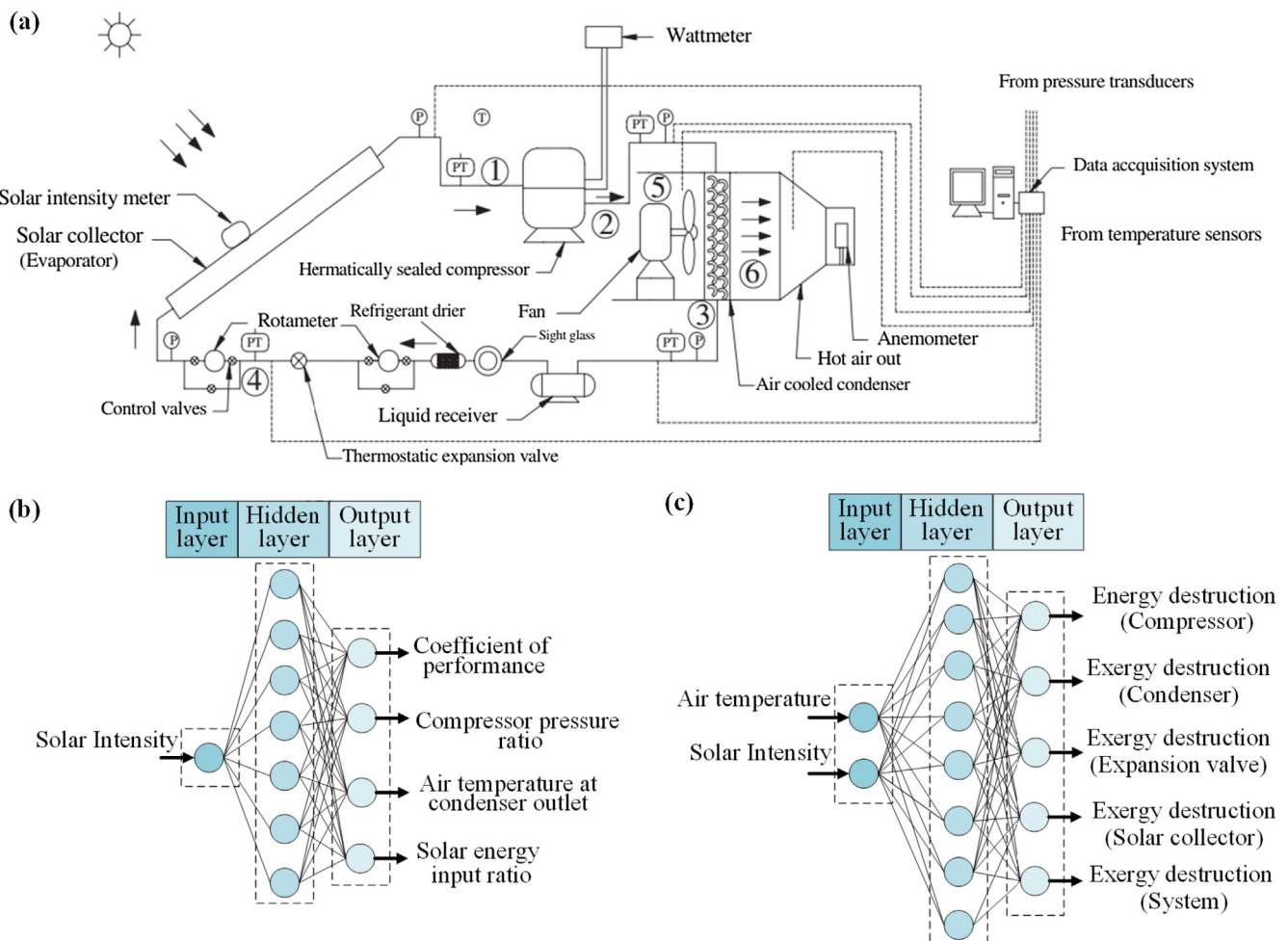


Fig. 7. (a) Schematic view of the experimental setup (Mohanraj et al., 2008; Mohanraj et al., 2009a,b); (b) Architecture of the ANN (Mohanraj et al., 2008); (c) Architecture of the ANN (Mohanraj et al., 2009a).

compressor discharge temperature, and energy performance ratio. BP with LM learning algorithm was used. The difference between the experimental and the predicted ANN results is minimized using GA. The obtained results revealed that the ANN integrated with GA shows better performance as well as consumes less execution time compared with the ANN.

Esen et al. (2017) constructed an ANN and ANFIS models to predict the performance of solar-assisted heat pump system. The ANN optimal parameters (the number of hidden layers, number of the neurons in the hidden layers, and activation functions) were selected based on the minimum ANN error using the correct regression and the coefficient of multiple determinations as statistical performance evaluation criteria. The obtained results revealed that ANFIS model had better performance compared with ANN model.

3.5. Photovoltaic/thermal (PV/T) systems

PV is used to perform direct conversion of SE (light) into electrical energy (Al-Waeli et al., 2017). However, the performance of PV system is highly affected by the temperature increase of the PV panel, as the efficiency decreases by 5% for every increase of 10 °C of the PV panel (Chow et al., 2003). Many researchers (Delisle and Kummert, 2014;

Nemati et al., 2016; Wu et al., 2017) have suggested integration of PV with a thermal system (PV/T systems) to remove this excessive heat using cooling fluids such as water, nanofluids, and air to enhance the overall efficiency of the system. It was found that PV/T systems has more advantages over PV systems and solar thermal systems. Many numerical analyses have been conducted for various PV/T systems to avoid the high cost of field tests by predicting outputs against some input variables (Al-Shamani et al., 2018; Ni et al., 2018; Paradis et al., 2017; Simonetti et al., 2018). ANNs are used for this purpose and showed a satisfactory prediction, especially for the non-linear nature of the training data (Al-Waeli et al., 2018).

Kalani et al. (2017) introduced a new method to model the relationship between input and output parameters of a photovoltaic system consists of collector system working by nanofluid as a working fluid using three different artificial intelligence methods, namely, MLP, RBF, and ANFIS. Particle Swarm Optimization (PSO) technique was used to optimize the structure of each model to overcome the non-linearity and complexity of the problem (Elsheikh and Abd Elaziz, 2018). While the outputs of MLP are calculated by linear combinations of the hidden layers neurons outputs which adjusted by a weighted average of the inputs using some transfer functions such as sigmoid function, the outputs of RBF are calculated based on mapping distances

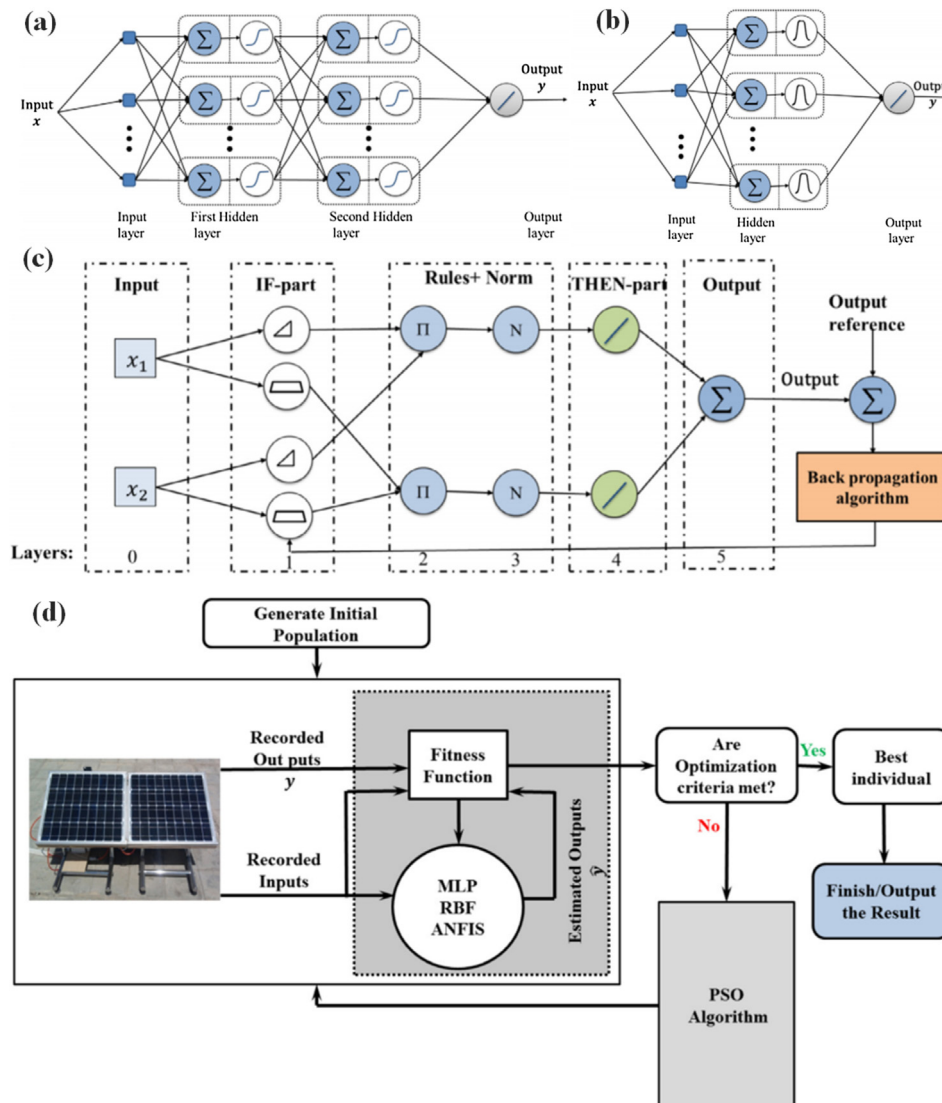


Fig. 8. (a) Structural diagram of MLPANN; (b) Structural diagram of RBFANN; (c) Structural diagram of ANFIS; (d) Block diagram of the proposed technique (Kalani et al., 2017).

between center vectors and input vectors and then the output is calculated through a nonlinear kernel. MLP can have more multiple hidden layers, while the number of the hidden layers of the RBF is restricted to one layer. BP training of both MLP and RBF were performed using Levenberg Marquardt algorithm, while ANFIS was learned by a hybrid algorithm consists of gradient descent and the least-squares method. Fig. 8(a)–(c) shows the structure of the models under study. Fig. 8(a) shows the architecture of the MLP model which consists of four layers (an input layer, two hidden layers, and an output layer). Fig. 8(b) shows the architecture of the RBF model which consists of three layers (an input layer, a hidden layer, and an output layer). Fig. 8(c) shows the architecture of the ANFIS model which composed of an input layer followed by an *if* block, *rules* block, and *then* block; and the outputs of the *then* block are summed up in the output layer. PSO was used to determine the best value of the prediction process parameters, due to its ability to find the global optimum with much faster convergence as well as its simplicity, and its high accuracy (Seyedmahmoudian et al., 2014). Then MLP, RBF, and ANFIS are used to determine the best mapping function to estimate the target outputs based on the PSO results as shown in Fig. 8(d). The inputs parameters for the three models are the incident solar radiation, ambient temperature, and fluid inlet temperature; while the outputs parameters are the working fluid outlet temperature and the electrical efficiency. ANFIS had the best performance in predicting the outputs parameters compared with the MLP and RBF. (Kamthania and Tiwari, 2012) used ANN to analyze the performance, including thermal energy, overall thermal energy, electrical energy and overall exergy, of a semi-transparent hybrid photovoltaic thermal double pass air collector. The ANN input parameters are ambient solar radiation intensity, air temperature, number of clear days and diffuse radiation. Log-sigmoid transfer function was applied to the neurons of the hidden layer. Gunasekar et al. (2015) developed ANN model for the prediction of the energy performance, including evaporator heat gain, SE input ratio, photovoltaic efficiency and photovoltaic panel temperature, of a PV/T evaporator used in solar assisted heat pumps. The ANN input parameters are ambient solar radiation intensity, ambient wind velocity, ambient air temperature, and ambient relative humidity. Multilayer feed forward network with BP learning algorithm, Levenberg-Marquardt training algorithm, one hidden layer, log-sigmoid transfer function was used. The number of neurons in the hidden layer was determined using the following formula (Kalogirou and Bojic, 2000):

$$\text{Number of hidden neurons} = \frac{1}{2}(\text{input neurons} + \text{output neurons}) + \sqrt{\text{Number of training data}} \quad (15)$$

Analysis of variance and various statistical performance evaluation criteria, i.e., RMSEs, a fraction of absolute variance values, and minimum COV values were used to assess the ANN performance. It was found that ambient temperature and solar intensity have the greatest effect on the energy performance of the system.

Ghani et al. (2012) used ANN to predict photovoltaic yield for a building integrated photovoltaic thermal array of specific mass flow rate and aspect ratio. Levenberg-Marquardt BP algorithm was used to train the ANN. Sigmoid non-linear activation function was used for the hidden layer and the output layer. The input parameters of the ANN were the mass flow rate and arrays aspect ratio. The output parameters of the ANN are the photovoltaic yield under four fluid flow configurations, namely single riser/reverse flow, single riser/parallel flow, double riser/ reverse flow, and double riser/parallel flow. Ravaee et al. (2012) used ANN to model the performance (thermal and electrical efficiencies) of a PV/T collector. The ANN input parameters are ambient temperature, the fluid temperature at duct inlet, cell temperature, solar intensity, fluid velocity in duct, and time. The Levenberg-Marquardt BP algorithm was used for the training process. Sigmoid and linear transfer functions were used for the hidden layer and the output

layer, respectively.

Karatepe et al. (2006) used an ANN approach for modeling a photovoltaic module. The input parameters of the ANN are solar radiation and ambient temperature, while the output parameters are the diode saturation current, ideality factor, series resistance, photo-current and parallel resistance. BP algorithm with Levenberg-Marquardt optimization method was used to update the neurons weights and biases. Hyperbolic tangent sigmoid and pure linear transfer functions were applied to activate of the hidden layer and output layer, respectively. The model is composed of two stages. Firstly, the ANN is used to predict the five aforementioned output parameters using the solar irradiance and ambient temperature as input parameters. Secondly, these five parameters are fed to one diode electrical equivalent circuit model to get the current and voltage of the module. Mellit et al. (2013) estimated the profile of the produced power of Si-polycrystalline photovoltaic (PV) module using two different ANNs for sunny and cloudy days, respectively. The ANN input parameters are PV module current and voltage, solar irradiance, and air temperature. Input data were measured for more than one year. Hyperbolic tangent sigmoid and linear transfer functions were used for the hidden and output layers, respectively. The ANN model showed better performance compared with multiple linear regression, polynomial regression, one-diode and analytical models. Ammar et al. (2013) constructed an ANN model to compute the optimum flow rate that allows an optimum operating point for a given solar radiation and ambient temperature of a PV/T system and to determine the optimal thermal and electrical power. The input parameters of the model were temperatures of the solar cell, absorber plate, glazing, and water circulation; while the outputs were the thermal profit and the electric power output.

In another study, Graditi et al. (2016) proposed a hybrid physical ANN to estimate the energy yield of a 1.05 kW which may be beneficial to avoid outages and to ensure quality and continuity of the supply. The results of the ANN-based model were compared with those obtained from physical-based modeling of the plant. The measured and estimated data from both approaches have been compared based on different statistical coefficients. The proposed approach allows a more precise estimation of the energy yield with RMSE less than 10%. Recently, Khatib et al. (2018) proposed two different ANN models (generalized regression ANN and cascaded forward ANN) to predict I–V for the different types of commercial solar cells under different solar radiation and temperature conditions. The inputs of the models are solar radiation, cell temperature, current, voltage, short circuit current, and open circuit current. The RMSE values are in the range of 0.013–0.073 A, which reveal the efficiency of the proposed model for predicting the I–V curve for different PV modules.

3.6. Solar cookers

Energy consumed during cooking accounts for 36% of the total global primary energy consumption (Bansal et al., 2014). Solar cooker which utilize SE to cook food, as an efficient alternative to traditional cooking devices, has a great importance in the domestic sector for saving energy (Herez et al., 2018; Nkhonjera et al., 2017; Yettou et al., 2014). Various designs of solar cookers for domestic applications, such as solar panel cookers, solar parabolic cookers, and solar box cookers, have been theoretically and experimentally investigated to optimize their performance (Geddami et al., 2015; Joshi and Jani, 2015; Mirdha and Dhariwal, 2008).

(Kurt et al., 2008) predicted the thermal performance parameters of a solar cooker (shown in Fig. 9(a)) such as pot water, enclosure air and absorber plate temperatures using a FFNN based on BP algorithm. The inputs of the ANN are solar radiation intensity, time of day, ambient temperature, water amount in the cooking pot, and solar cooker type (with or without reflector). The proposed ANN architecture, including input, hidden, and output layers, as well as input and output parameters are shown in Fig. 9(b). The logistic sigmoid transfer function was

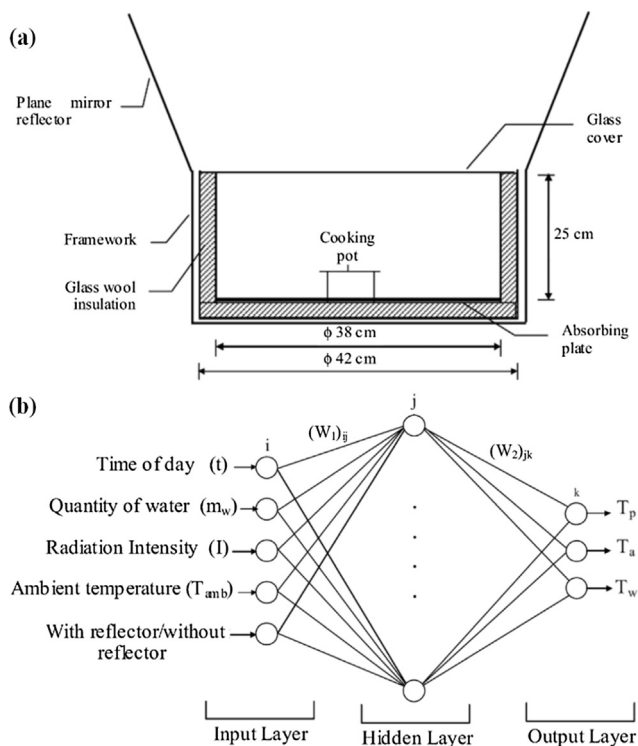


Fig. 9. (a) Cross section of experimental solar cooker; (b) The architecture of the proposed ANN (Kurt et al., 2008).

applied. The values of ANN's parameters such as the number of neurons, momentum coefficient, and learning rate are determined using minimum sum squared errors. The predicted temperatures of solar cooker have a RE within $\pm 10\%$ compared with the exact measured temperatures.

3.7. Greenhouses

Greenhouses are used to enhance the yield per cultivated area up to 10 times compared with those of free land cultivation, however, they suffer from their high fossil fuels consumption. They are classified as highly non-linear dynamic and complex coupled systems as their performance is affected by many variables such as ambient temperature, wind velocity, moistening cycle and humidity. Many modeling methods have been used to control greenhouses such as transfer function, physical and black-box modeling. ANN proved to be one of the best modeling techniques used to overcome non-linearity, strong coupling, and time-invariant of the greenhouse system. Taki et al. (2016) compared ANN with some mathematical based models (dynamic and multiple linear regression) to predict energy transfer, roof temperature and inside air temperature and in a semi-solar greenhouse shown in Fig. 10(a). The environment factors (ambient temperature, soil temperature, solar radiation, wind speed and inside air humidity) that influencing the predicted values were collected as data samples. Gradient Descent Momentum and LM were used in the training process. Sigmoid and linear transfer functions were used for the hidden layer and the output layer, respectively. MLP showed a better performance compared with mathematical based models. Castañeda-Miranda and Castaño (2017) used MLP, trained by Levenberg-Marquardt BP algorithm, to control greenhouses under frost conditions. Ambient temperature, solar radiation, inside and outside air relative humidity, and wind speed were used as input parameters in the ANN to predict the air temperature inside the greenhouse.

Fourati (2014) proposed ANN with an Elman structure (a type of recurrent NNs) and composed of an input layer, an output layer, a

context layer, and a hidden layer (Fig. 10(c)) to model the greenhouse. While the two former layers interact with the environment, the two later layers do not. Fig. 4(d) shows the learning algorithm of the Elman NN to represent the greenhouse's direct dynamics. Then adaptive resonance theory NN was exploited to decompose the greenhouse database into sub-clusters database to get different local models of the greenhouse, and to select the appropriate neural controller based on the operation mode (Fig. 10(b)). The system inputs are ambient temperature, solar radiation, ambient hygrometry, wind speed, thermal power, opening, shade, sprayer; while the outputs are internal hygrometry, and internal temperature. The obtained results indicate that the error between the calculated outputs and the desired outputs has been reduced using multiple neural control scheme compared with that of the single neural control strategy.

He et al. (2007) introduced a hybrid BPNN integrated with GA to model a greenhouse system. The output of the system are the temperature and humidity of the inside air. The inputs are ambient temperature, solar radiation, relative humidity, and wind speed. GA was used to avoid the trapping of NN to a local optimum and to increase learning rate and convergence speed. NN was improved by adding the self-adaptation learning rate and inertia impulse to increase the learning speed and decrease convergence vibration. Then the thresholds of NN and initialized weights were optimized using a GA. In another study, Zarei and Behyad (2019) developed an ANN model to examine the effect of design parameters of a solar seawater greenhouse, in which solar energy is utilized to distillate seawater and humidify the inner surface of the greenhouse, on the fresh water yield. The input parameters of the model were length, width, roof transparency, and the height of the front evaporator. Among all investigated algorithms that used to train the ANN, such as conjugate gradient algorithm, the gradient descent algorithm, Bayesian, and the LM algorithm, LM showed the best performance with RMSE of 1.136 l.

3.8. Solar dryers

Solar dryers are used as practical, eco-friendly and economical technology in agricultural field to preserve seeds, crops, vegetables, and fruits (Mustayen et al., 2014; Prakash et al., 2016; Singh Chauhan et al., 2015). Bala et al. (2005) used MLP approach trained using BP algorithm to estimate the performance of the solar tunnel drier (Fig. 11(a)) for drying leather and bulbs of jackfruit. The drier consists of a tunnel drying unit integrated with a flat-plate solar air collector covered by a UV-stabilized transparent plastic and connected to a small fan operated by a photovoltaic module used to supply air over the dried fruit. The ANN consists of four layers, namely: input layer, two hidden layers, and output layer. The input variables of the ANN are solar radiation intensity, ambient temperature, initial moisture content, relative humidity, airflow rate, drying time, and drying object (leather or bulb); while the output variable is the final moisture content as shown in (Fig. 11(b)).

Tripathy and Kumar (2009) developed and evaluated ANN model to predict the temperature of food product (potato cylinders and slices) in a solar dryer. The input parameters of the ANN are the solar radiation and the ambient temperature. Several training algorithms (LM BP, Polak–Ribiere conjugate gradient back propagation, SCG back propagation, BFGS quasi-Newton BP, and Resilient BP) as well as various transfer functions (log sigmoid, saturating linear, hyperbolic tangent sigmoid, and positive linear) were examined to select the optimal NN configuration. Based on minimum measures of error between the measured and predicted data, log sigmoid transfer function with Resilient BP algorithm is the most appropriate training approach for both potato cylinders and slices. Çakmak and Yıldız (2011) applied feed-forward NN to model the nonlinear behavior of the drying process of seedy grapes. A drying room, with swirl flow drying mechanism using SE collected by a solar air collector with phase-change material and an expanded-surface solar air collector, was developed. The ANN model

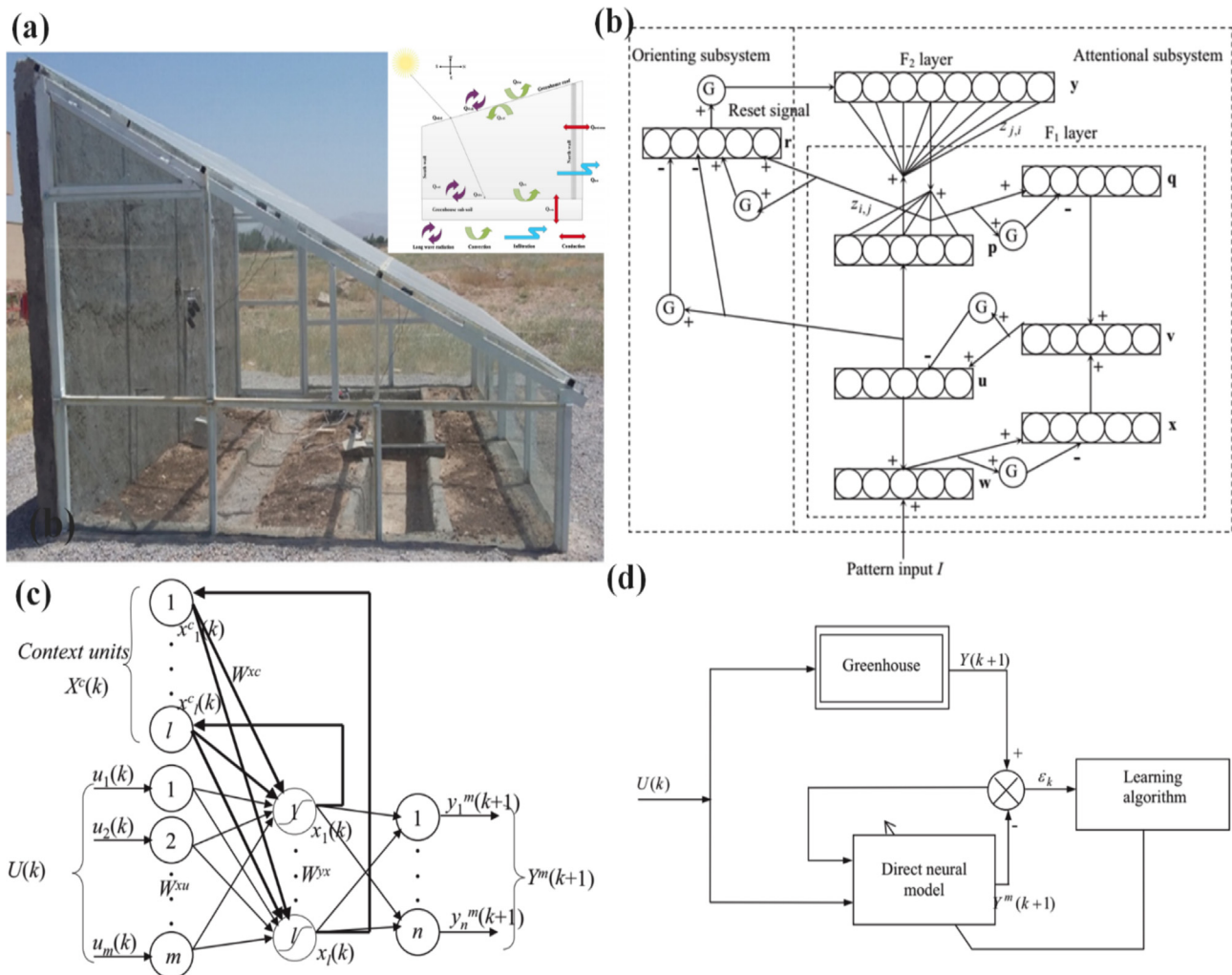


Fig. 10. (a) picture of semi-solar greenhouse (Taki et al., 2016); (b) Elman neural network structure; (c) Direct neuronal model learning process; (d) adaptive resonance theory NN (Fourati, 2014).

performed more consistently and was more accurate compared with linear and nonlinear regression models when their performance was compared by means of the MAEs, the root means square errors and the CC statistics. The ANN input parameters are the hot air flow rate, the hot air temperature difference, and the moisture content, while the ANN output is the drying rate. The ANN with LM optimization method with hyperbolic tangent sigmoid and linear transfer functions for the hidden layer and output layer, respectively, showed the best performance.

3.9. Solar stills

A solar still is a very simple device powered by the solar radiation and used in water desalination by utilizing simultaneous evaporation and condensation processes (Edalatpour et al., 2016; Sharshir et al., 2017a; Vishwanath Kumar et al., 2015). The brine or impure water inside the solar still is heated up and evaporated by SE passing through a glass cover at which the evaporated water condensate again and collected as the distilled water output. There are many parameters that may affect the productivity of solar stills such as the area of absorption, water depth, glass cover cooling, inlet water temperature, phase change materials, and meteorological parameters (Elsheikh et al., 2019; Sharshir et al., 2018; Sharshir et al., 2017b; Sharshir et al., 2017c).

(Mashaly, A.F. and Alazba, A., 2016; Mashaly, A.F. and Alazba, A.A., 2016; Mashaly and Alazba, 2017; Mashaly et al., 2015) modeled

the water productivity, instantaneous thermal efficiency, and operational recovery ratio of a solar still (shown in Fig. 12(a)) using two different methods. The first method is multiple linear regressions (MLR) as a well-known based statistical method used to model the relationship among multiple variables linearly. The second method is MLP as an artificial intelligence based method which capable of representing complex and nonlinear input/output relationships. The MLP model is composed of one input layer, one hidden layer, and one output layer as shown in Fig. 12(b). In this study, only one hidden layer is used as increasing the number of hidden layers may result in the problem of local minima (Rai et al., 2005). Both sigmoid function and hyperbolic tangent transfer functions were examined for the hidden layer, however, the later showed better statistical parameters values. The input parameters of the MLP are Julian day, solar radiation, ambient temperature, ultra violet index, wind speed, relative humidity, the temperature of brine water, the temperature of feed water, and total dissolved solids of brine water and feed water. MLP model showed better results compared with MLR based on different performance evaluation criteria, including the COD, performance index, the OIMP, index of agreement, RMSE, the coefficient of residual mass, fractional bias, and model efficiency. The relative errors of the predicted instantaneous thermal efficiency using MLP method were within $\pm 10\%$. In another study, (Mashaly and Alazba, 2015) compared between three different ANN learning algorithms, namely, the LM, the resilient BP, and the conjugate gradient BP with Fletcher–Reeves restarts. The LM algorithm

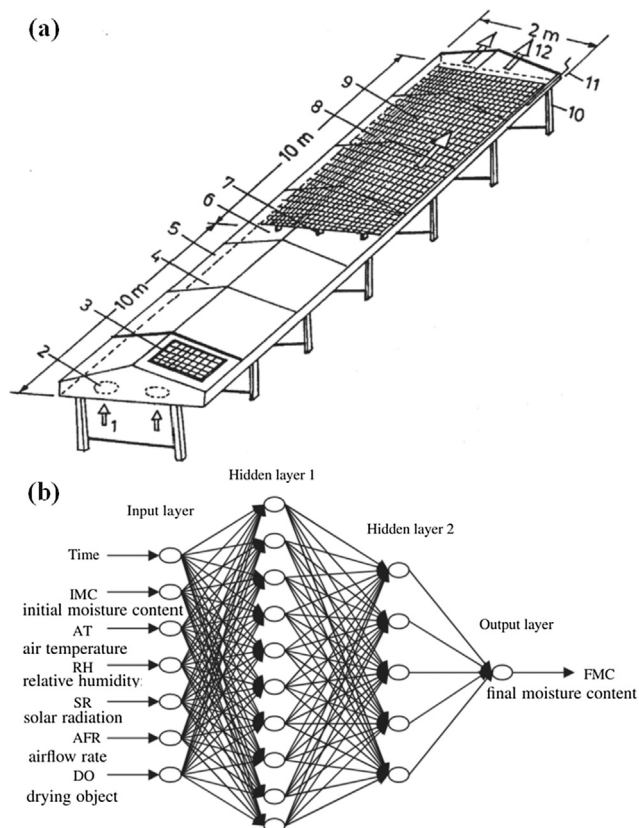


Fig. 11. (a) solar tunnel drier; (b) the structure of ANN used in (Bala et al., 2005).

was found to be the best in the training, validation, and testing stages based on OIMP and root mean squared error as statistical performance evaluation criteria.

Santos et al. (2012) predicted solar still distillate water productivity of two different commercial solar stills operated for eighteen months using ANN. The inputs parameters of the ANN model are the daily total insolation, daily average wind velocity, daily average wind direction, daily average cloud cover and daily average ambient temperature. The hidden layer consisted of twenty neurons with tangent sigmoid function. All input variables were normalized between zero and 1 to enhance the NN's generalization capabilities and to accelerate the training process. The normalization was done by dividing by the maximum value in each variable's range. The relative errors of the predicted yield using the ANN method were within $\pm 10\%$ depending on the input variables.

4. Conclusion and future work

In this paper, applications of ANNs for modeling of different SE devices have been reviewed. Previous studies reported potential advantages of the ANNs in modeling these devices, such as high accuracy, generalization capabilities, and short computing time, over other theoretical and experimental modeling techniques. The use of ANNs avoids solving complicated mathematical models despite of the simplification assumptions used in the ANN-based modeling. Moreover, it requires less experimental tests to figure out the input/output relationships compared with experimental studies. Therefore, the ANN-based modeling of SE systems may save time and reduce the financial expenses. MLP neural network has been widely used in SE application due its simplicity. The Levenberge-Marquardt learning algorithm, in most cases, has proven to be flexible and robust with fast optimization compared with other learning algorithms. Further studies with more

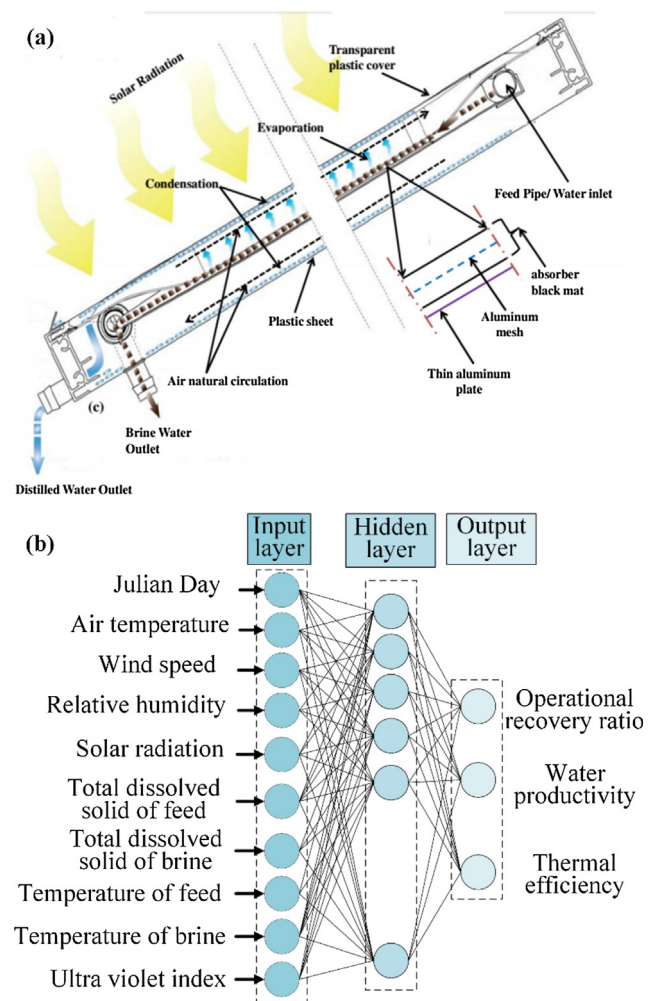


Fig. 12. (a) cross sectional view of the solar still panel; (b) Architecture of the MLP NN model used to predict the instantaneous thermal efficiency (Mashaly, A.F. and Alazba, A.A., 2016; Mashaly et al., 2015).

focus on other types of ANNs such as recurrent neural networks and ANFIS should be carried out. Moreover, there exist a trend in machine learning to use the metaheuristic methods such as genetic algorithms (GA) and particle swarm optimization (PSO), grey wolf optimization (GWO), sine cosine algorithm (SCA) to improve the performance of the ANNs. Since, these methods have several properties such as they are global optimization methods which avoid stuck at local solutions. Also, most of them are population-based methods which means they consider more than one solution to the problem (the weights or parameters of ANN). Based on those properties, these global methods can find the optimal weights between the input-hidden layers and hidden-output layers. Therefore, this hybrid between ANNs and metaheuristic optimization methods can achieve high accuracy when applied to different SE applications. In addition, the other extensions of the ANNs such as ANFIS, extreme machine learning (EML), and deep neural networks have high quality than traditional ANNs in different applications. So, it constructs a new trend to use them in SE applications. In addition, the other extensions of the ANNs such as ANFIS, extreme machine learning (EML), and deep neural networks have high quality than traditional ANNs in different applications. So, it constructs a new trend to use them in SE applications.

Acknowledgment

This work was supported by the National Natural Science

Foundation of China (E050902, E041604).

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